AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 18 · STUDENT LESSON

Retrieval Augmented Generation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 17: distinguish retrieving evidence from phrasing an answer.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of building a grounded answer. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of building a **grounded answer**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

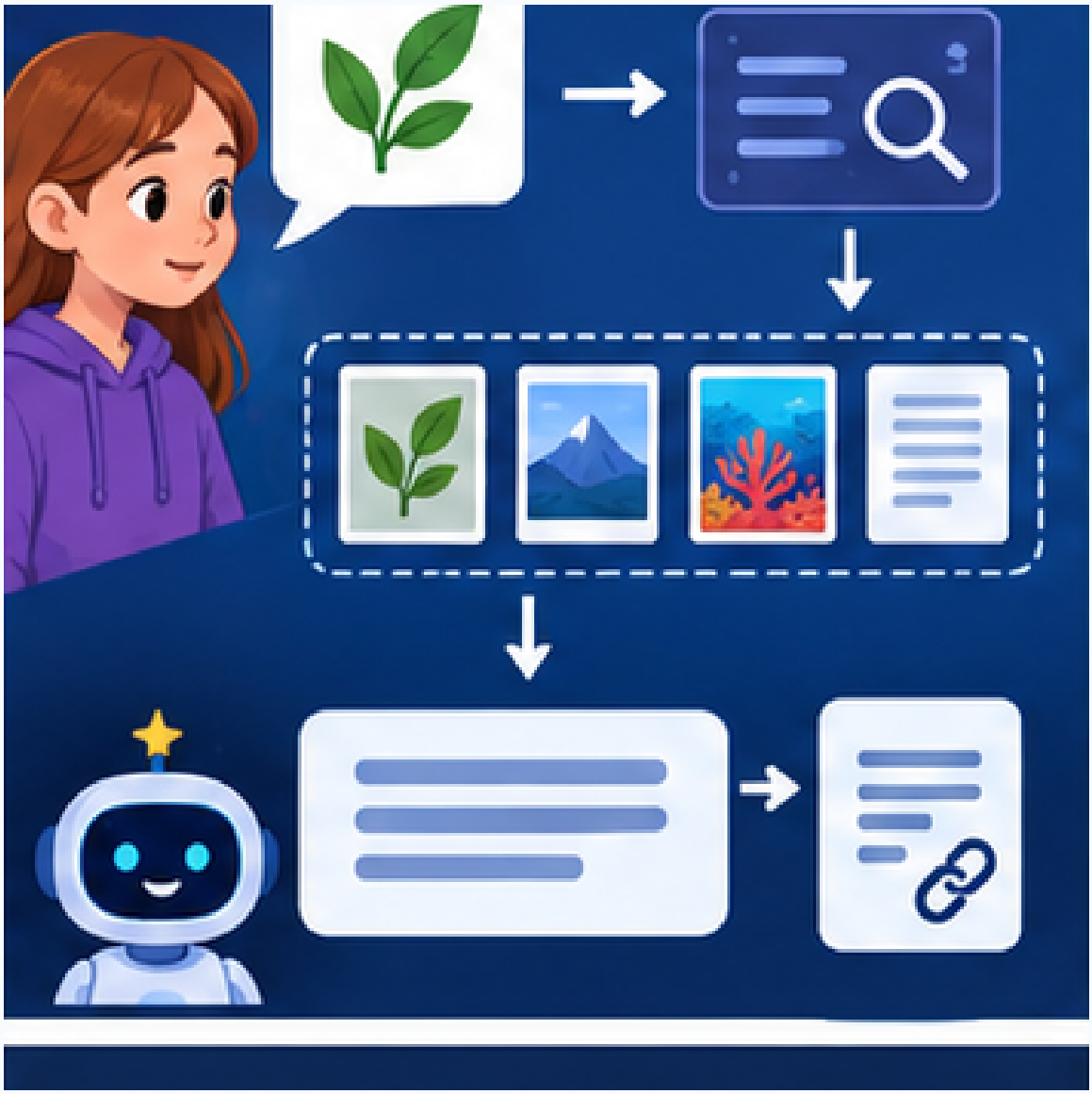
Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 18: Retrieval-Augmented Generation

RAG retrieves passages, then generates an answer grounded in them.



CREATOR MISSION

Trace **query** → retrieve → select → answer → cite.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Ranking and retrieval evaluation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

Readiness for Chunking and document structure

Recall Retrieval and indexing. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My readiness explanation

Readiness for Lexical and vector retrieval

Recall **Chunk**ing and document structure. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My readiness explanation

This week's end goal

By the end, I can explain retrieval augmented generation using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

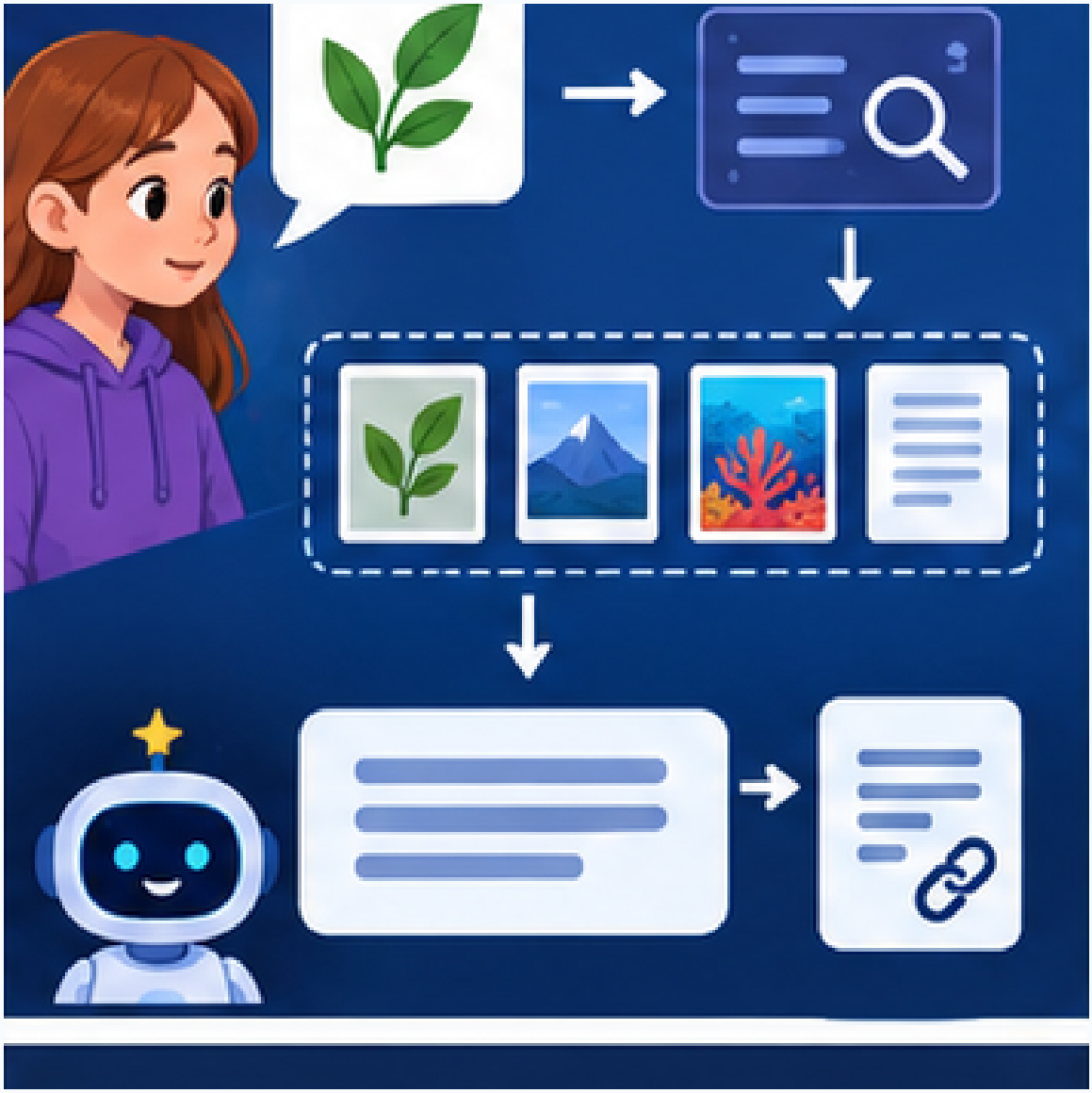
Notice the evidence

> Investigate the supplied Retrieval Augmented Generation case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use retrieval-augmented generation to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Chunking and document structure

Chunking divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Lexical and vector retrieval

Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

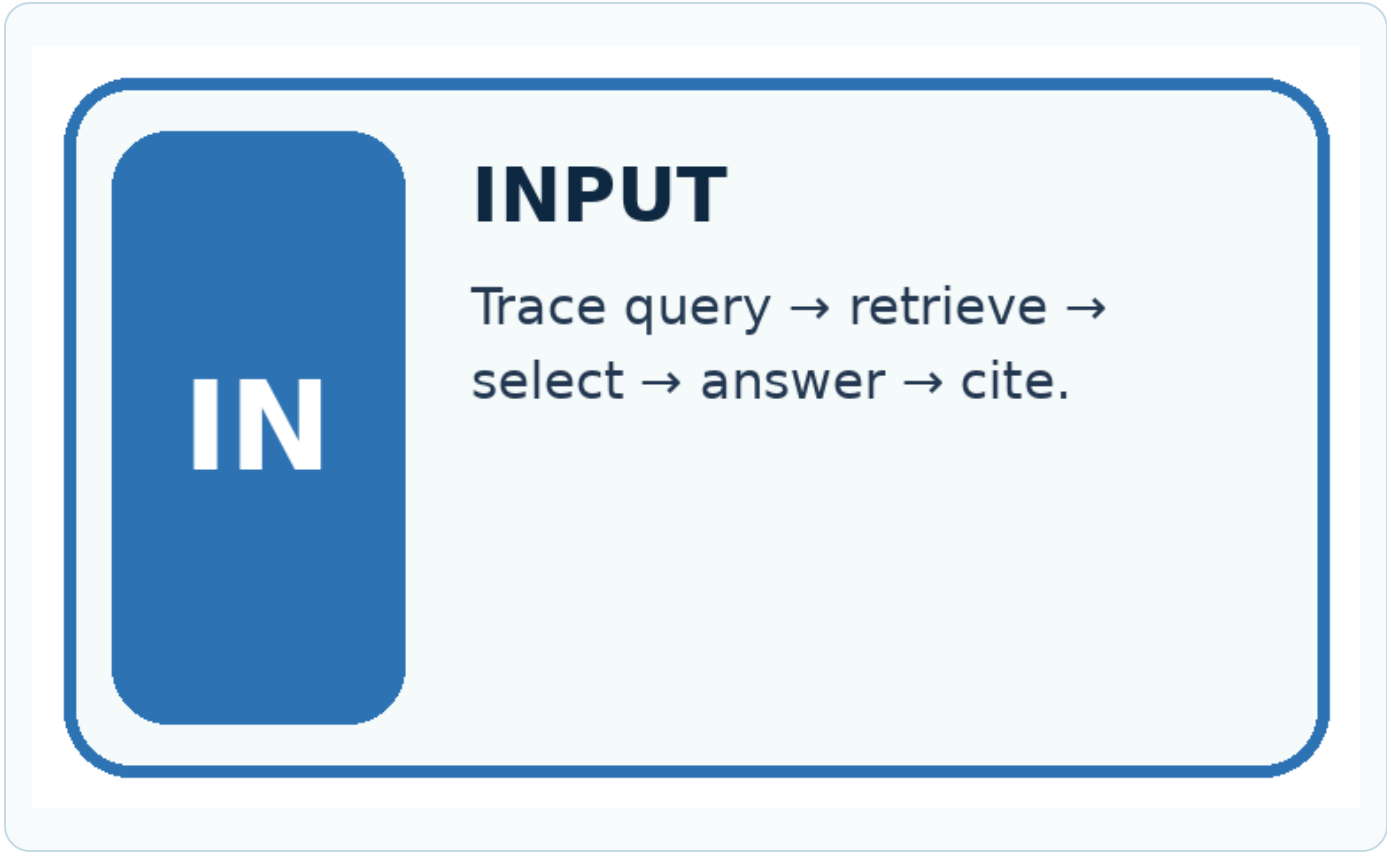
See how it works

> Follow the mechanism for Retrieval Augmented Generation. Connect each labelled input, deciding step and result.

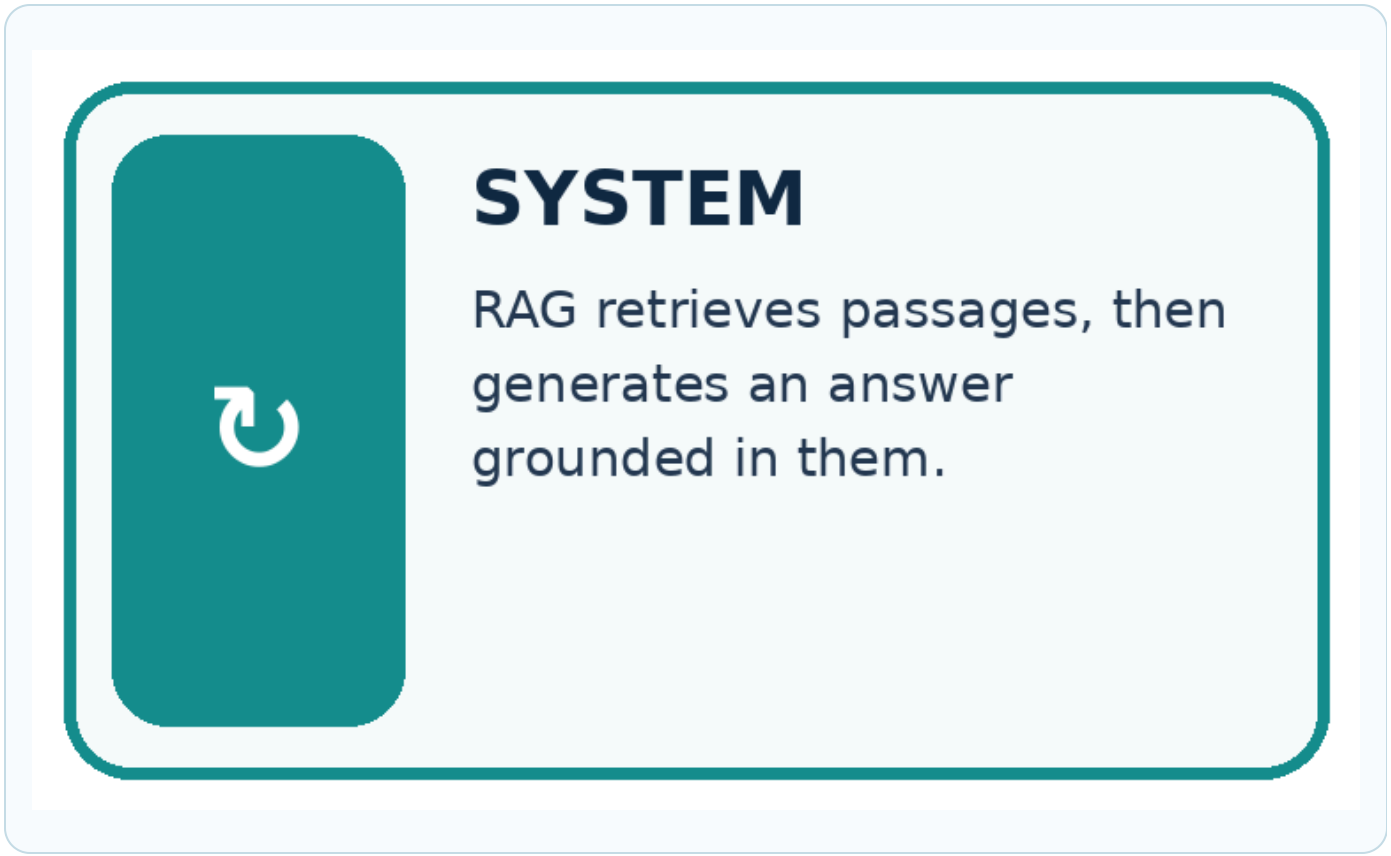
VISUAL MECHANISM

Input → Process → Output

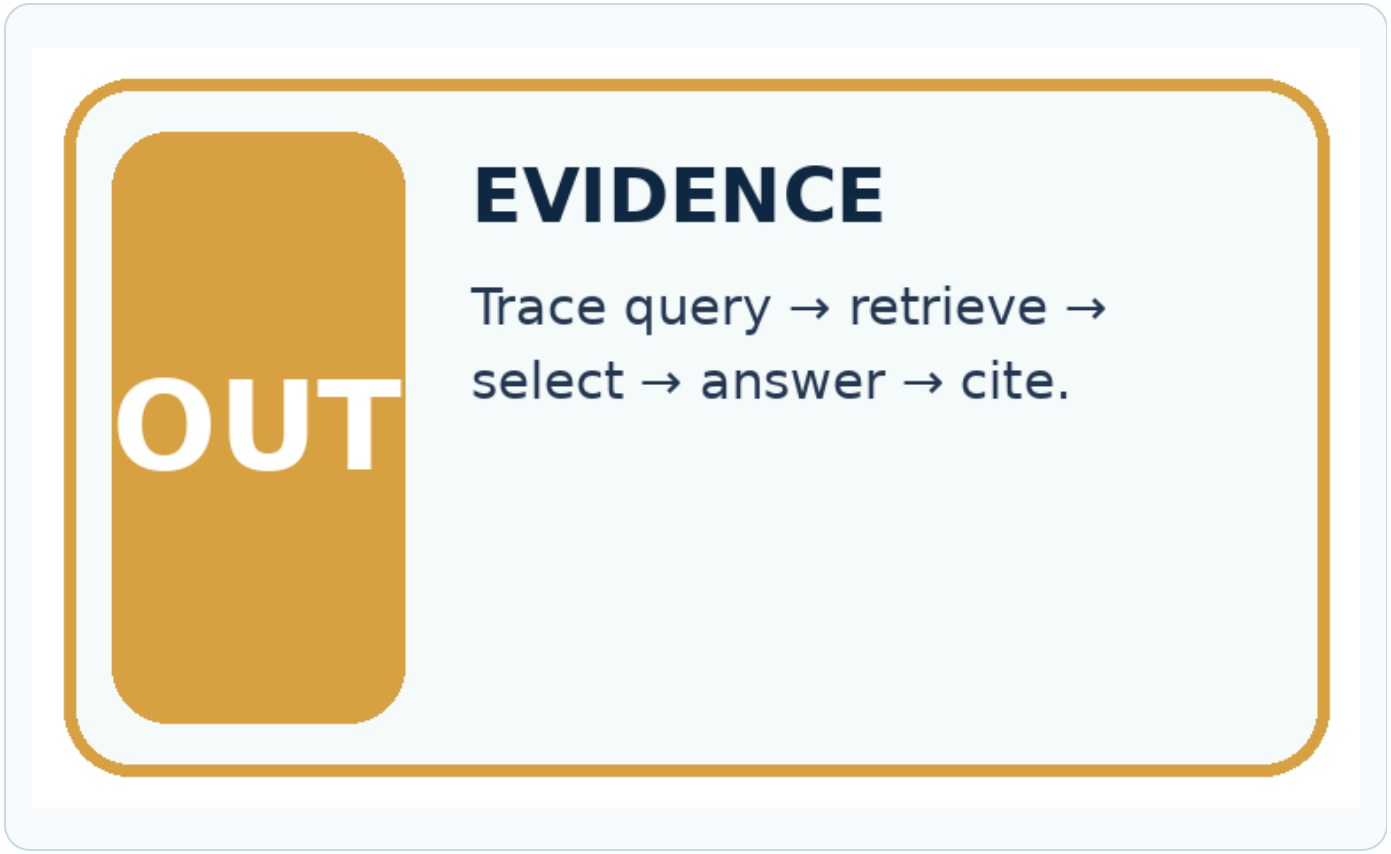
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Trace query → retrieve → select → answer → cite.

↓


SYSTEM
RAG retrieves passages, then generates an answer grounded in them.

↓



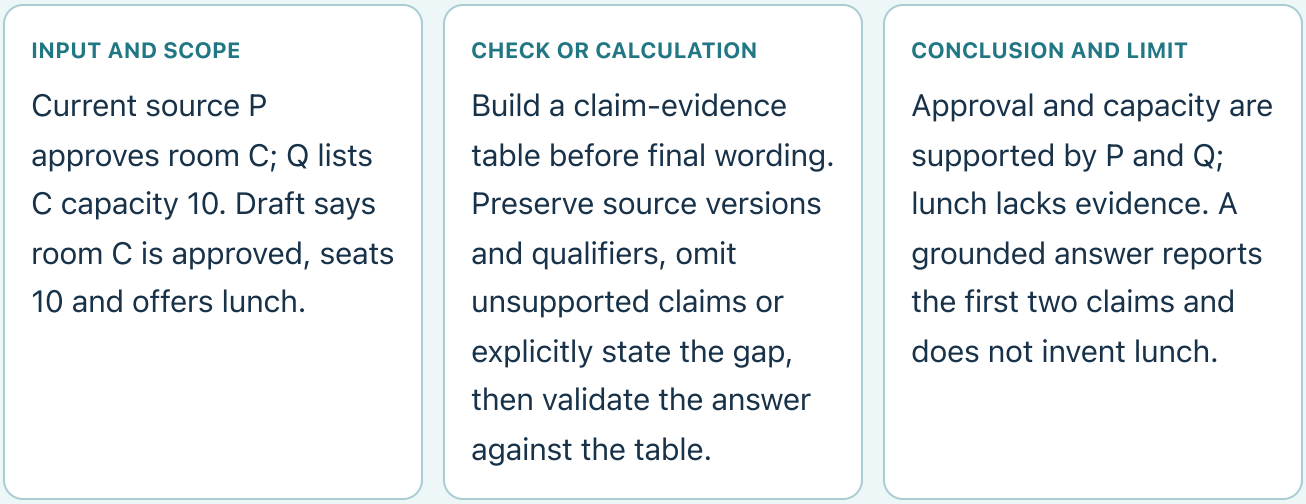
Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Build a claim-**evidence table** before final wording. Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

Building a grounded answer — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

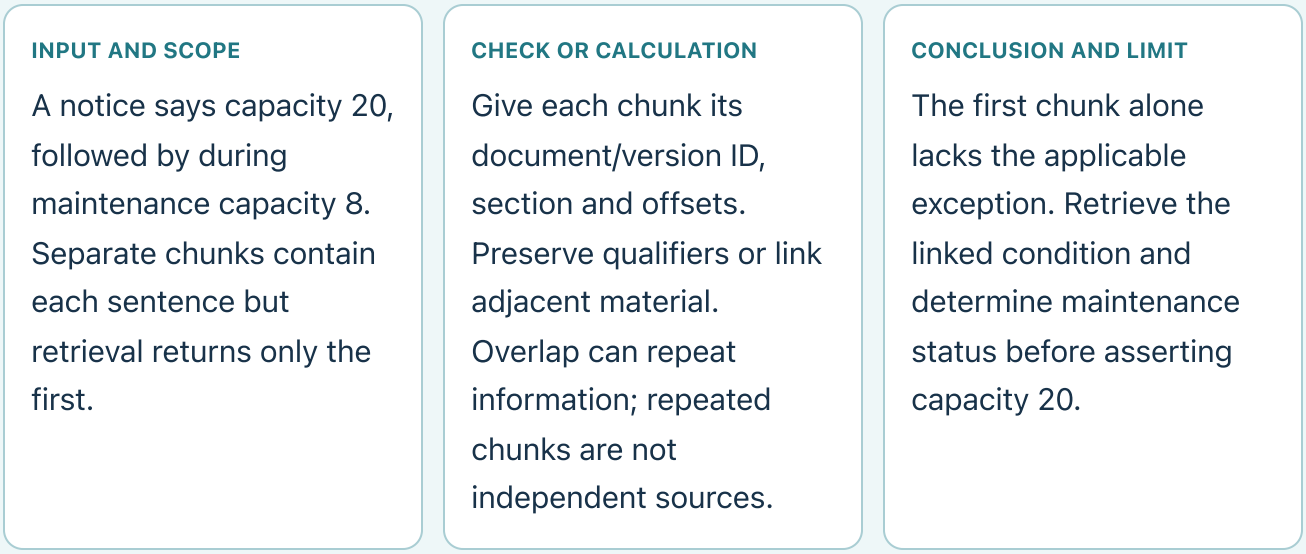
Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Chunking and document structure

Give each **chunk** its document/version ID, section and **offset**s. Preserve **qualifier**s or link adjacent material. **Overlap** can repeat information; repeated **chunk**s are not independent sources.

Chunking and document structure — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Lexical and vector retrieval

For a toy lexical score, count distinct **query** terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare **relevance** using independent judgments, not score magnitude alone.

Lexical and vector retrieval — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Query terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector query [1, 0], D1=[.4, .9], D2=[.8, .1].	For a toy lexical score, count distinct query terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare relevance using independent judgments, not score magnitude alone.	Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task relevance favours the sensor manual; vector ranking alone does not prove relevance.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

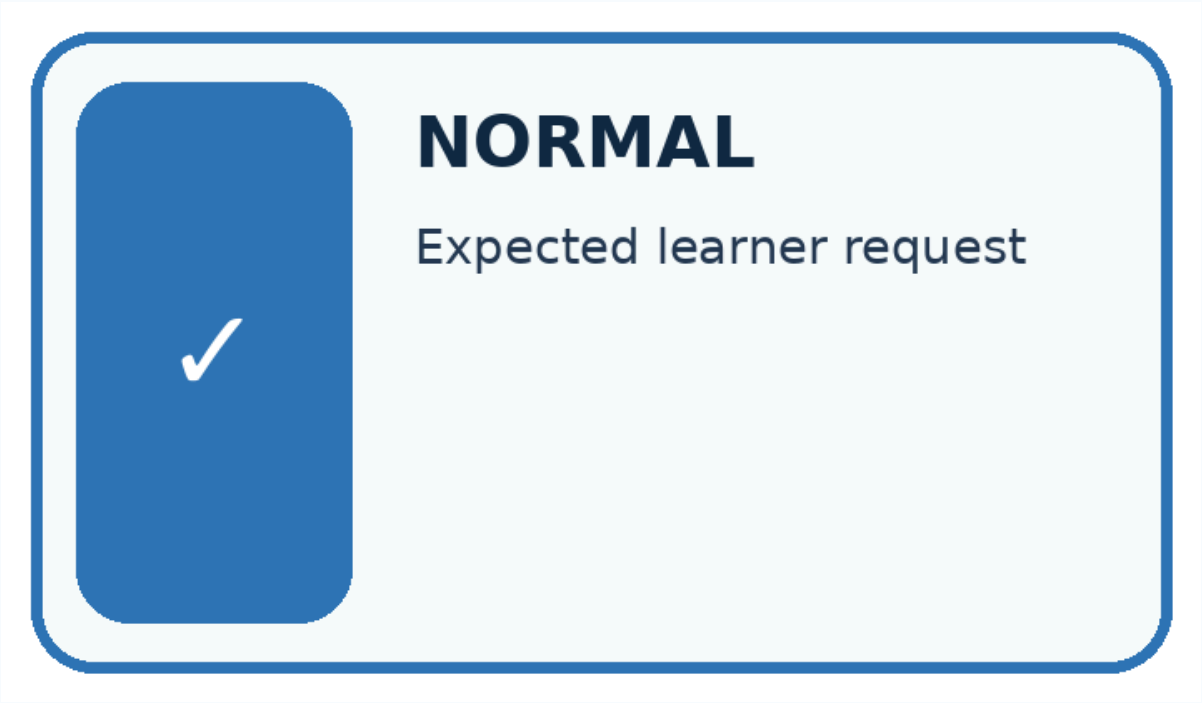
> Compare the supplied cases for Retrieval Augmented Generation. Identify the detail that changes the conclusion.

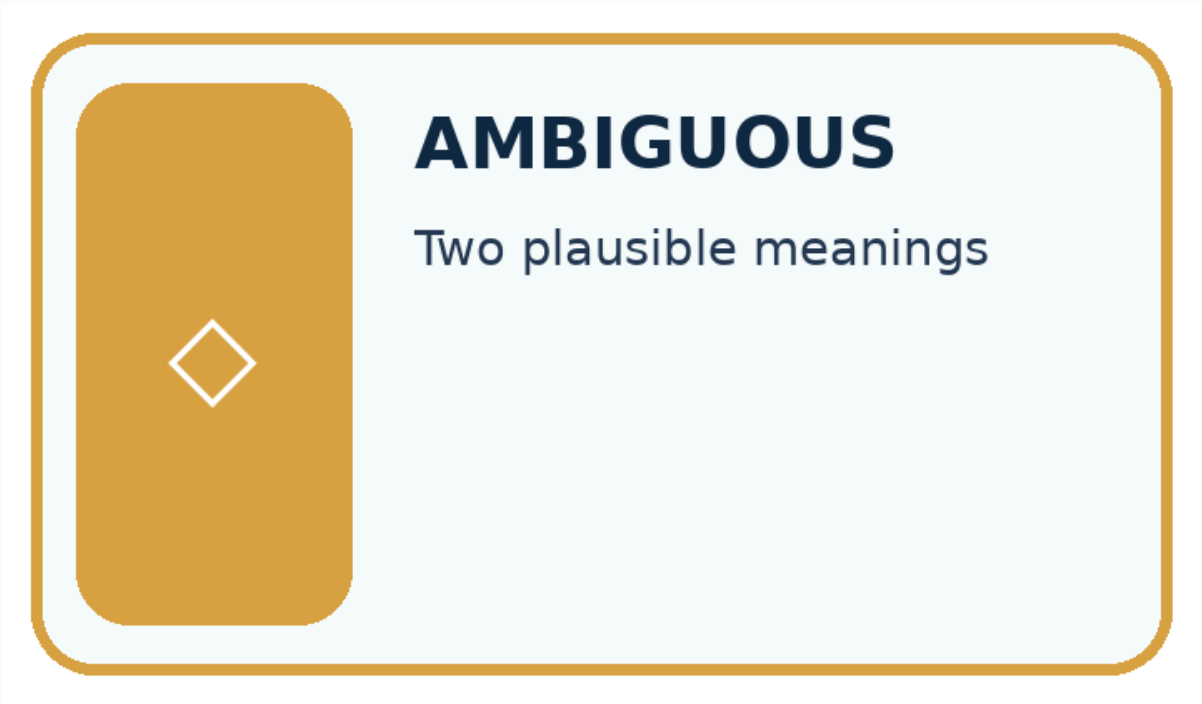
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

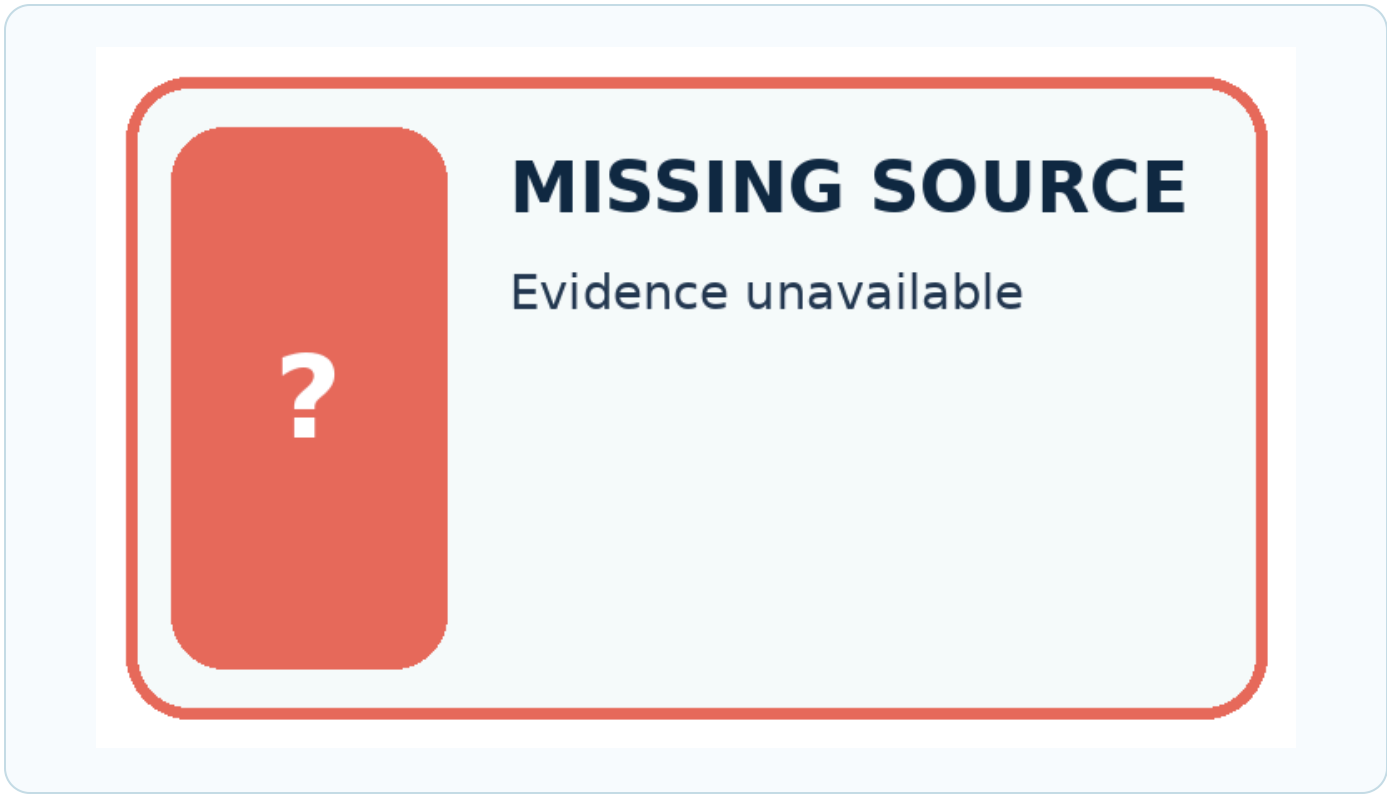
A system is not understood until it survives more than one comfortable example.

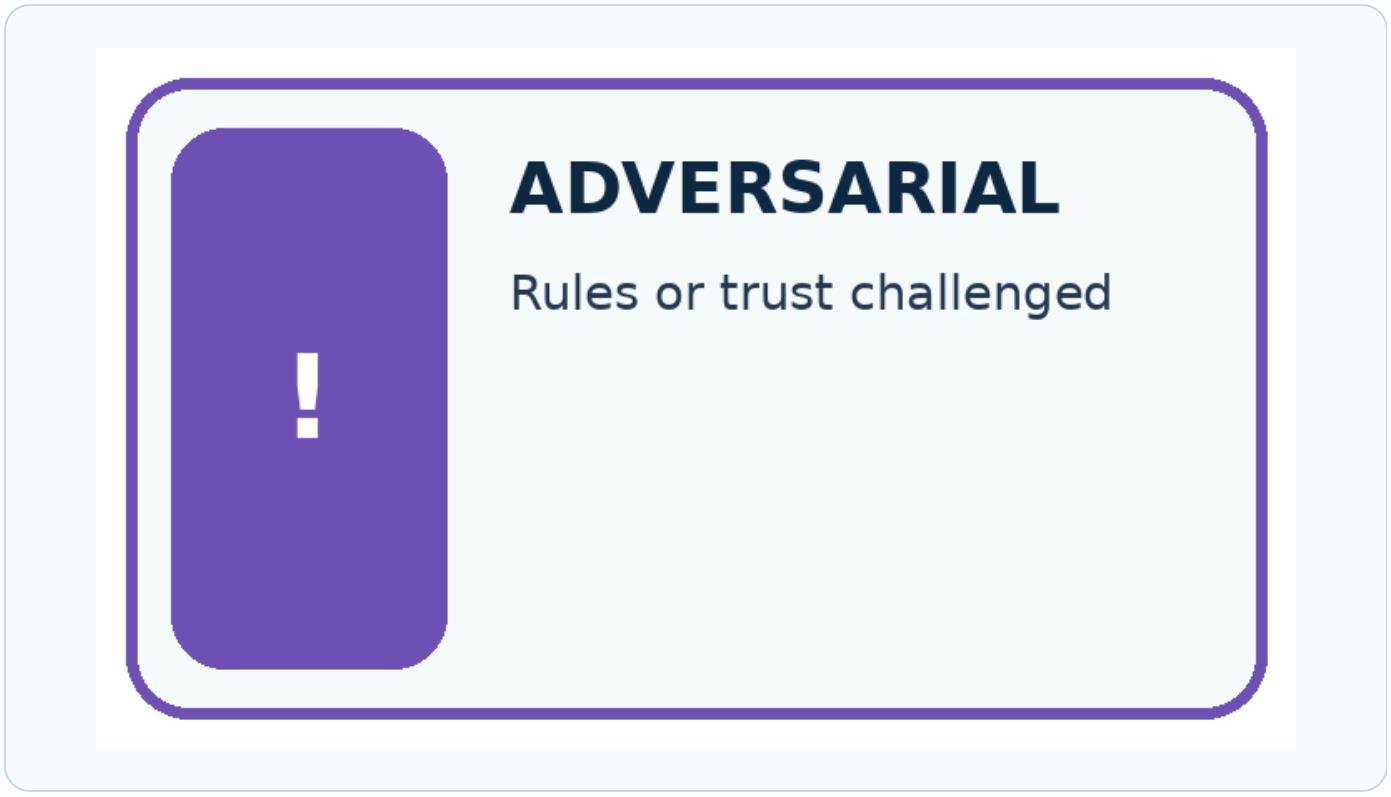




The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING SOURCE
Evidence unavailable


!
ADVERSARIAL
Rules or trust challenged

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: The question asks exactly how long this ice takes to melt.

Predicted result

Prediction: Say that S4 does not provide a time.

Why the result follows

Reason: A general change of state does not determine a duration for an unspecified sample.

Words that unlock the idea

> Use the vocabulary of Retrieval Augmented Generation to explain a concrete decision, rather than only memorising definitions.

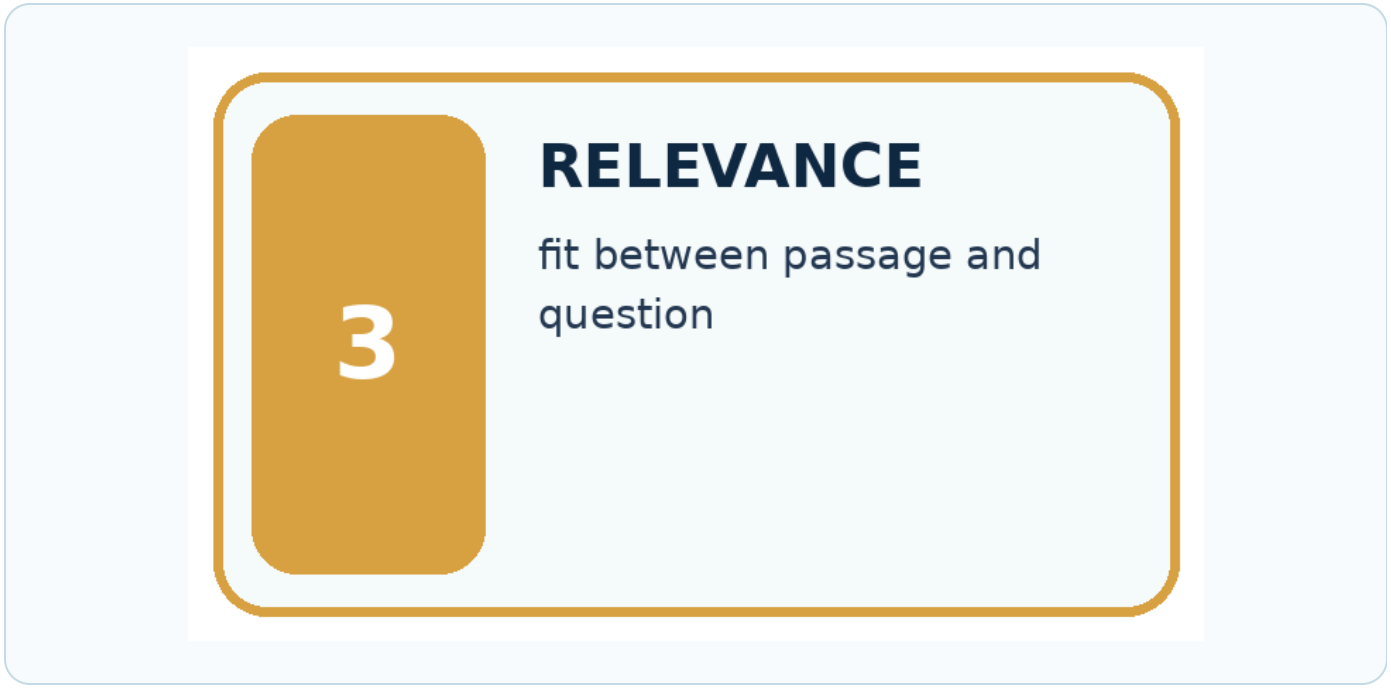
VOCABULARY LAB

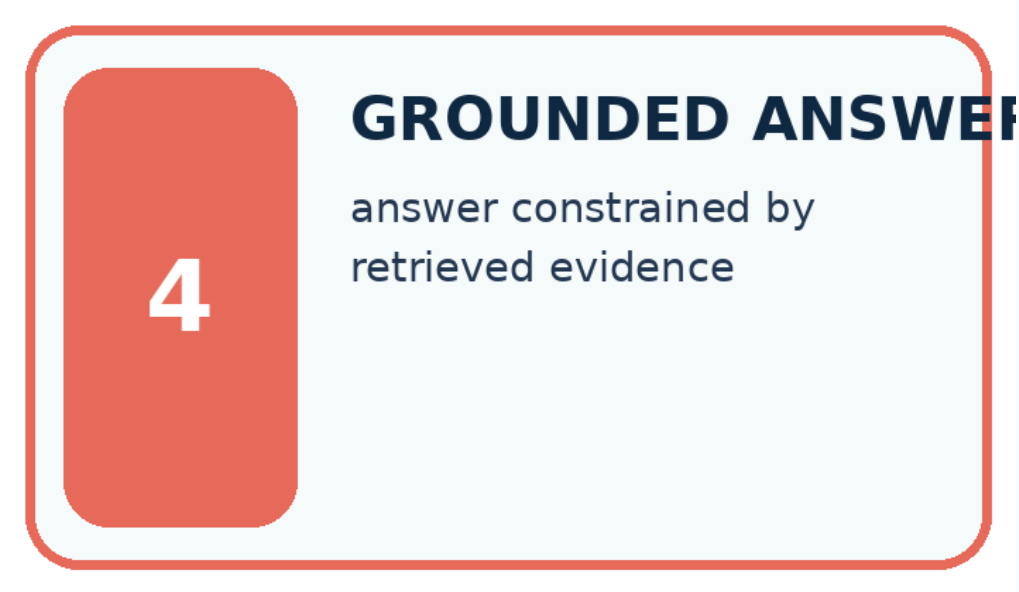
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
RETRIEVER
finds candidate passages


3
RELEVANCE
fit between passage and question



Word	My own example	Not this / boundary
query	A question or search request used to obtain information.	
retriever	The component that searches for and selects source material relevant to a query.	
relevance	How directly information helps answer the stated question or perform the task.	
grounded answer	An answer whose factual claims are supported by identified evidence relevant to the question.	

Words used in the closer explanation

Word	Meaning
Grounded claim	A claim supported by the available task evidence
Qualifier	A condition restricting a statement
Evidence table	A mapping from each claim to its supporting record

Terms for Chunking and document structure

Term	Meaning
Chunk	A retrievable unit from a document
Overlap	Content repeated in adjacent chunks
Offset	A location linking a chunk to the original source

Terms for Lexical and vector retrieval

Term	Meaning
Lexical match	Matching query terms to stored text
Vector retrieval	Ranking numerical representations using a declared similarity
Relevance judgment	An assessment of usefulness for the particular query

---

STUDENT LESSON · 6 OF 18

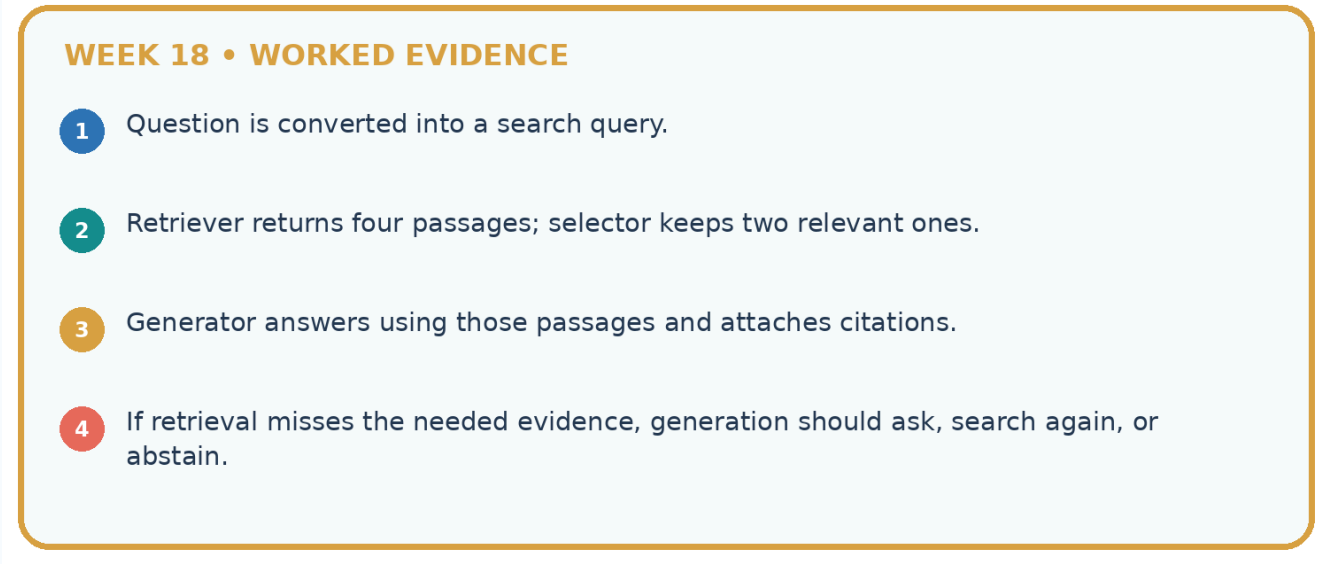
Follow a worked example

> Follow the worked trace for Retrieval Augmented Generation, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Question is converted into a search **query**.

2. **Retriever** returns four passages; selector keeps two relevant ones.

3. Generator answers using those passages and attaches citations.

4. If retrieval misses the needed **evidence**, generation should ask, search again, or abstain.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch.

Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Worked explanation · Retrieval Augmented Generation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Chunking and document structure

A notice says capacity 20, followed by during maintenance capacity 8. Separate **chunk**s contain each sentence but retrieval returns only the first.

The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

Worked explanation · Retrieval Augmented Generation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Lexical and vector retrieval

Query terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector **query** [1, 0], D1=[.4, .9], D2=[.8, .1].

Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

Worked explanation · Retrieval Augmented Generation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Retrieval Augmented Generation

Turn a short science source into an answer with a visible claim-to-source map.

The facts we will use

Supplied facts

Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/18

Retrieval Augmented Generation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read source

S4 supports the material identity and the solid-to-liquid change.



2. Draft two claims

Draft: ice is solid water; warming can turn it into liquid water.



3. Check each claim

Both claims match S4. A claim that melting takes exactly two minutes would not match.

CHANGE ONE THING

The question asks exactly how long this ice takes to melt.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read source	S4 supports the material identity and the solid-to-liquid change.
2. Draft two claims	Draft: ice is solid water; warming can turn it into liquid water.
3. Check each claim	Both claims match S4. A claim that melting takes exactly two minutes would not match.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

Keep the stated limit

Read the labelled picture: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The question asks exactly how long this ice takes to melt. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Retrieval Augmented Generation claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: RAG guarantees correctness because it has documents. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Chunking and document structure

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Lexical and vector retrieval

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

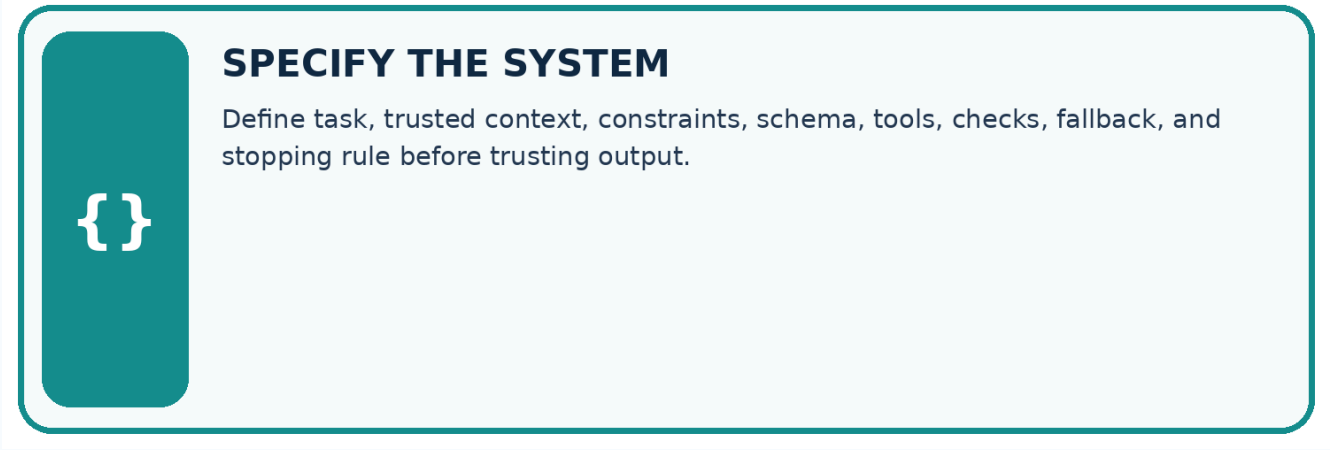
Practise together

> Practise Retrieval Augmented Generation with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Apply the connected idea: Chunking and document structure

Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Apply the connected idea: Lexical and vector retrieval

Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Building a **grounded answer**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Building a **grounded answer**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Chunking and document structure

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Practise the connected topic: Lexical and vector retrieval

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

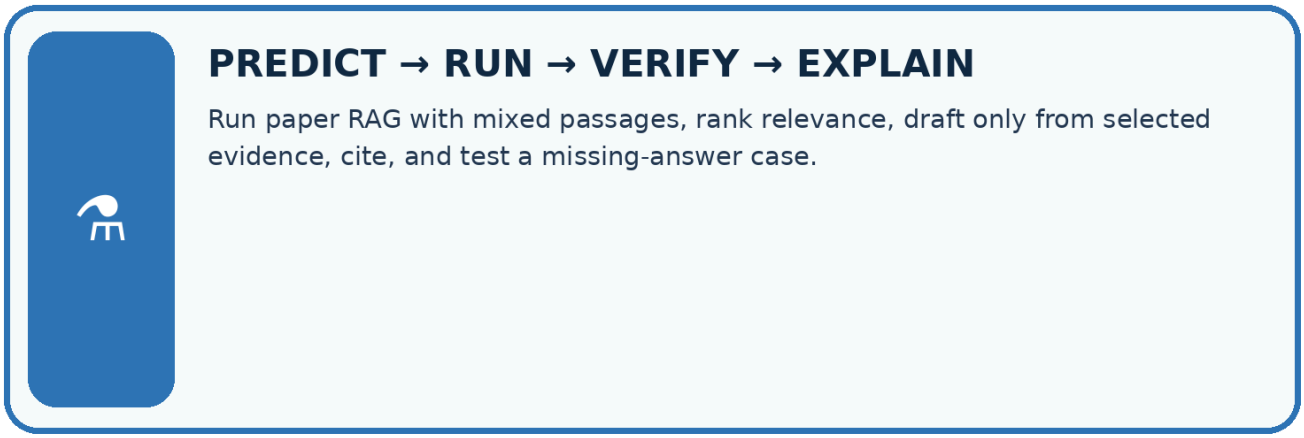
Run the investigation

> Investigate Retrieval Augmented Generation. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Run paper RAG with mixed passages, rank **relevance**, draft only from selected **evidence**, cite, and test a missing-answer case.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

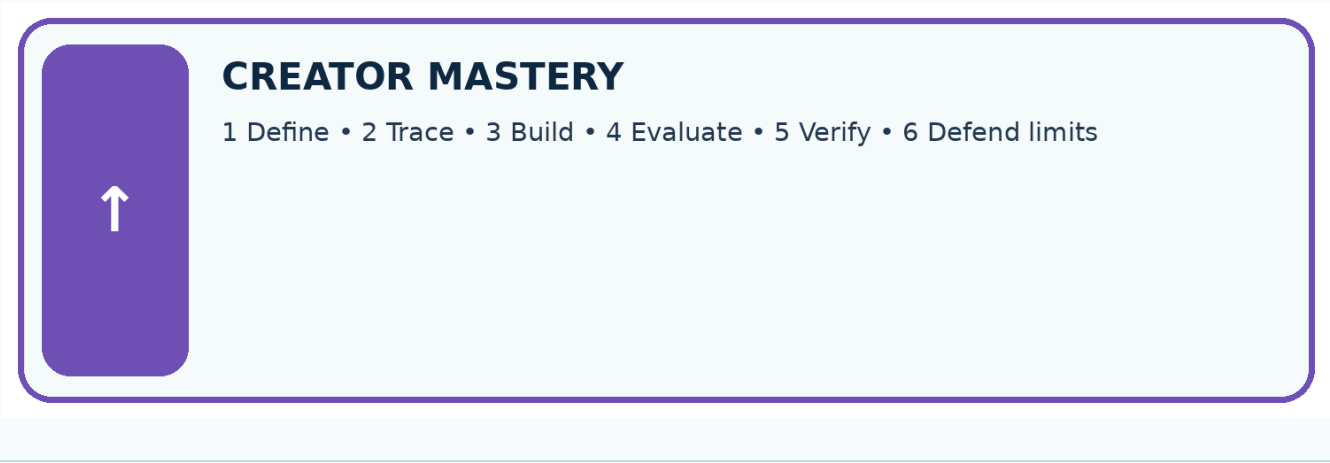
Build your understanding

Move from recall to tracing, diagnosis and transfer in Retrieval Augmented Generation. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

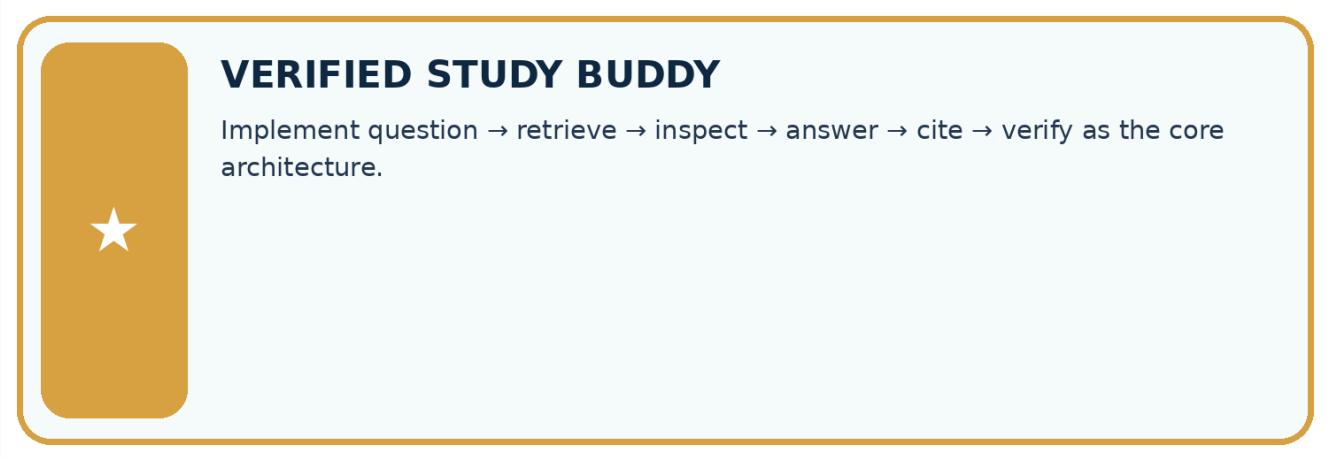
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Implement question → retrieve → inspect → answer → cite → verify as the core architecture.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Show what you know

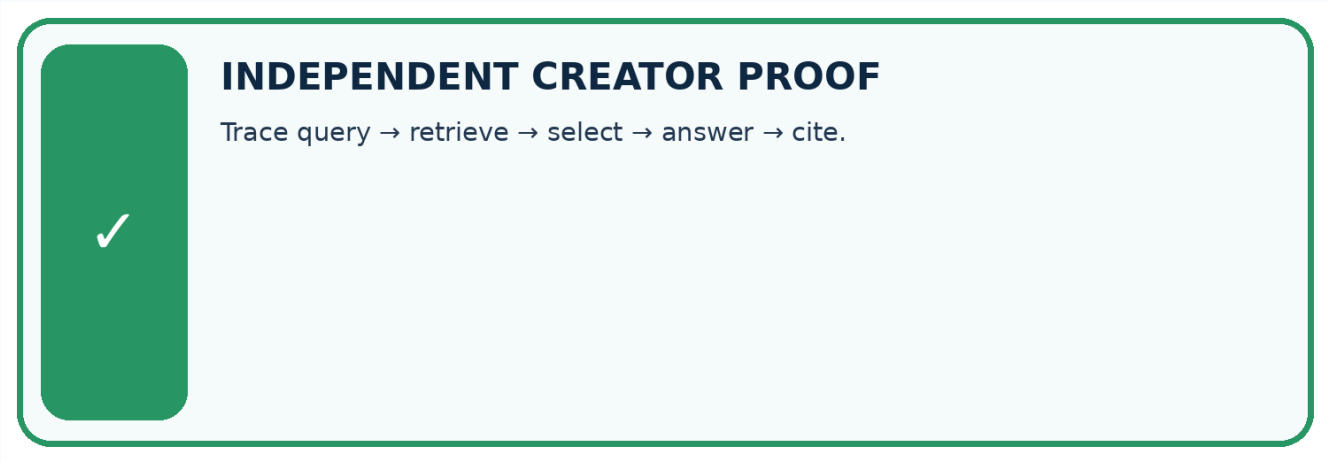
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Trace **query** → retrieve → select → answer → cite.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

Study a complete case

> Revisit the complete Retrieval Augmented Generation case for supported practice. Repeated examples do not become fresh assessments.

Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Note A says library opens at 9. Note B says returns go in the blue box.

The question

Answer opening time with **evidence**.

Worked explanation

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Use the passage that answers the question

Evidence or stage	Value or result
Question	When does the library open?
Retrieve	Note A says opens at 9
Compose	Library opens at 9
Verify	Note A supports opening time

Note A says library opens at 9. Note B says returns go in the blue box.

Answer opening time with **evidence**.

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A retrieved passage says the club meets Tuesday; it says nothing about cost. The assistant answers Tuesday, free entry. Which part is grounded and what should replace the unsupported part?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Implement question → retrieve → inspect → answer → cite → verify as the core architecture.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Sources: A='Library opens 9'; B='Cafe opens 8'; C='Library closes 5'. A question asks when the library opens. The **retriever** returns only B and C. What should the generator do?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Retrieval Augmented Generation, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Turn a short science source into an answer with a visible claim-to-source map.

By the end, I can	How I will show it
Explain the mechanism and terminology of building a grounded answer	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Retrieval Augmented Generation without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of building a **grounded answer**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Turn a short science source into an answer with a visible claim-to-source map.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both claims match S4. A claim that melting takes exactly two minutes would not match.

1. Check each claim

2. Draft two claims

3. Read source

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check each claim Both claims match S4. A claim that melting takes exactly two minutes would not match. Read the picture in order. Both claims match S4.

A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The question asks exactly how long this ice takes to melt. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Say that S4 does not provide a time.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Say that S4 does not provide a time. A general change of state does not determine a duration for an unspecified sample.

Repair a missing building block

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Review Speech and Audio Systems (ai-3/12). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?. Return to this week and explain what you changed.

Keep your completed work

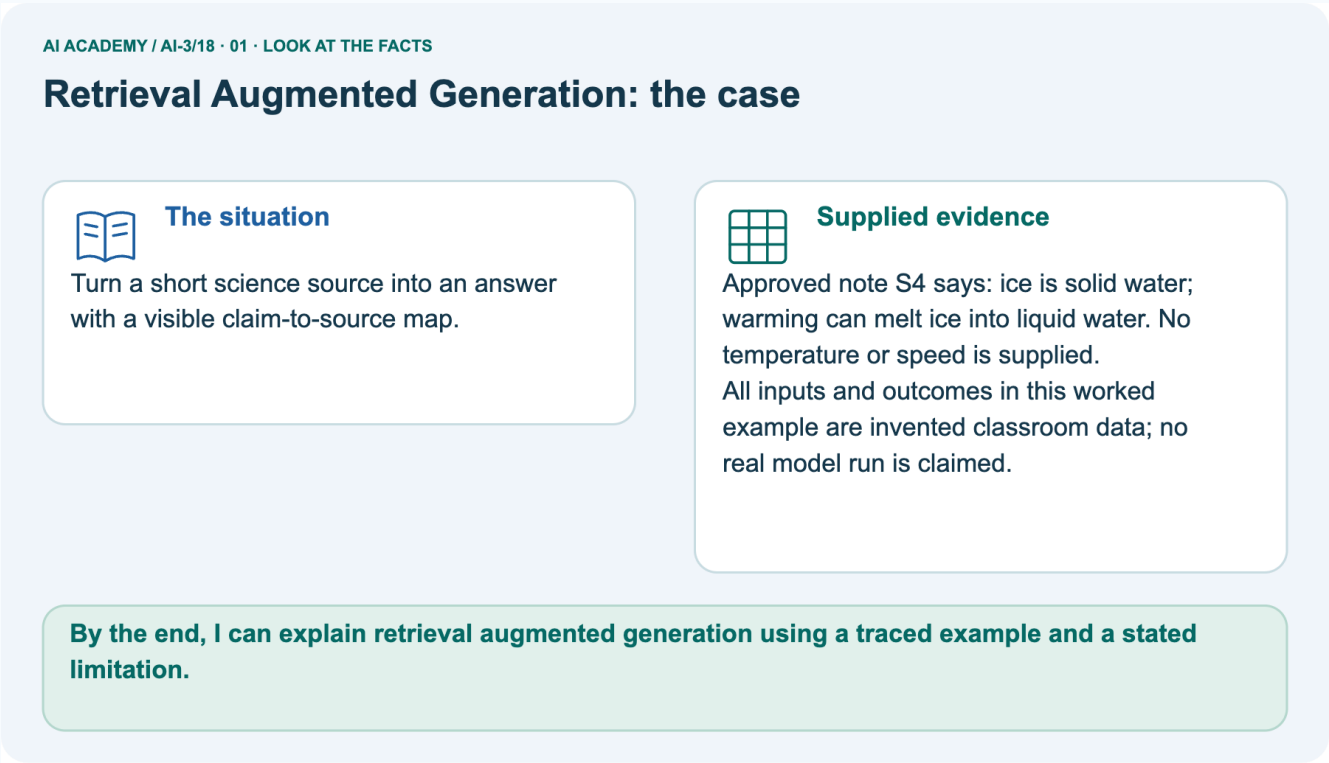
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Retrieval Augmented Generation graphic using its labels, deciding evidence and relevant limitation.

01. Retrieval Augmented Generation: the case



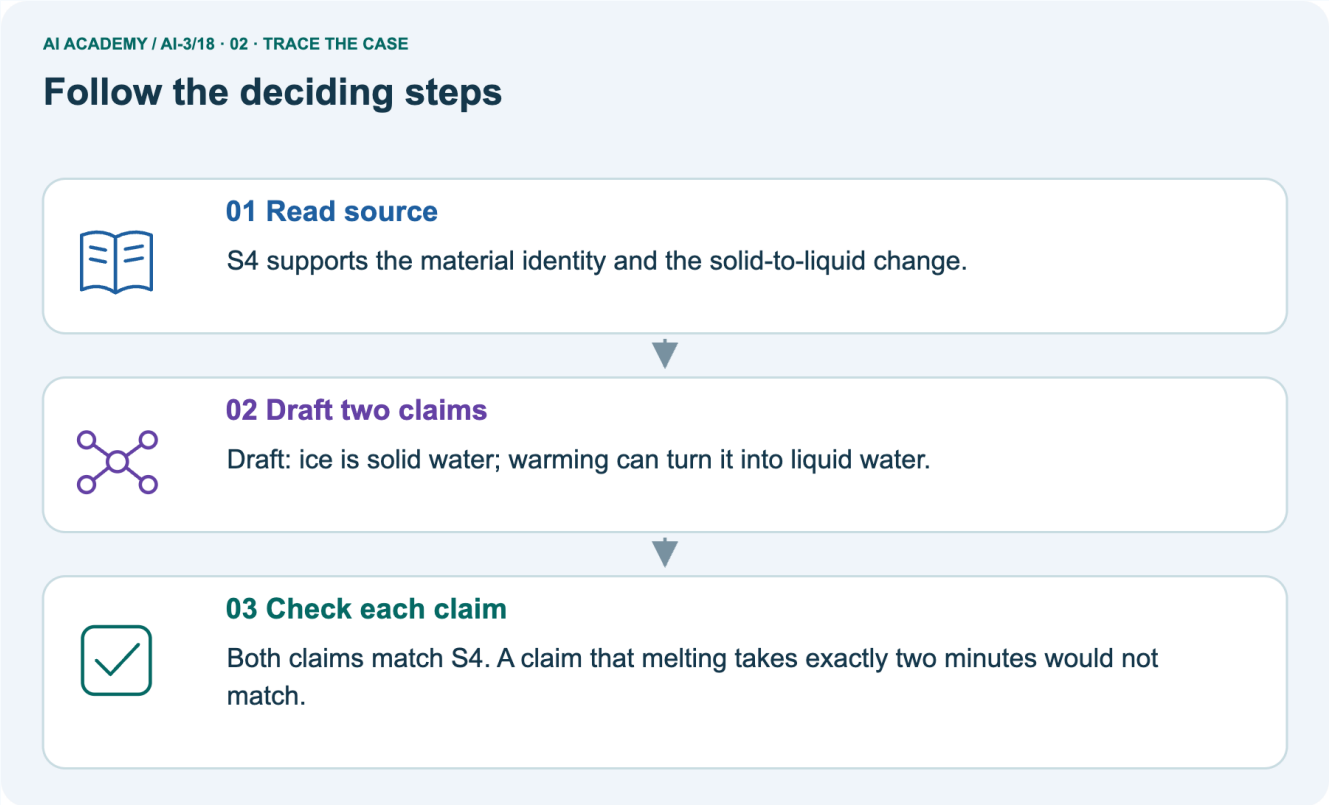
Follow the labelled picture with the matching explanation. Open the vector diagram

Turn a short science source into an answer with a visible claim-to-source map. Supplied facts: Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



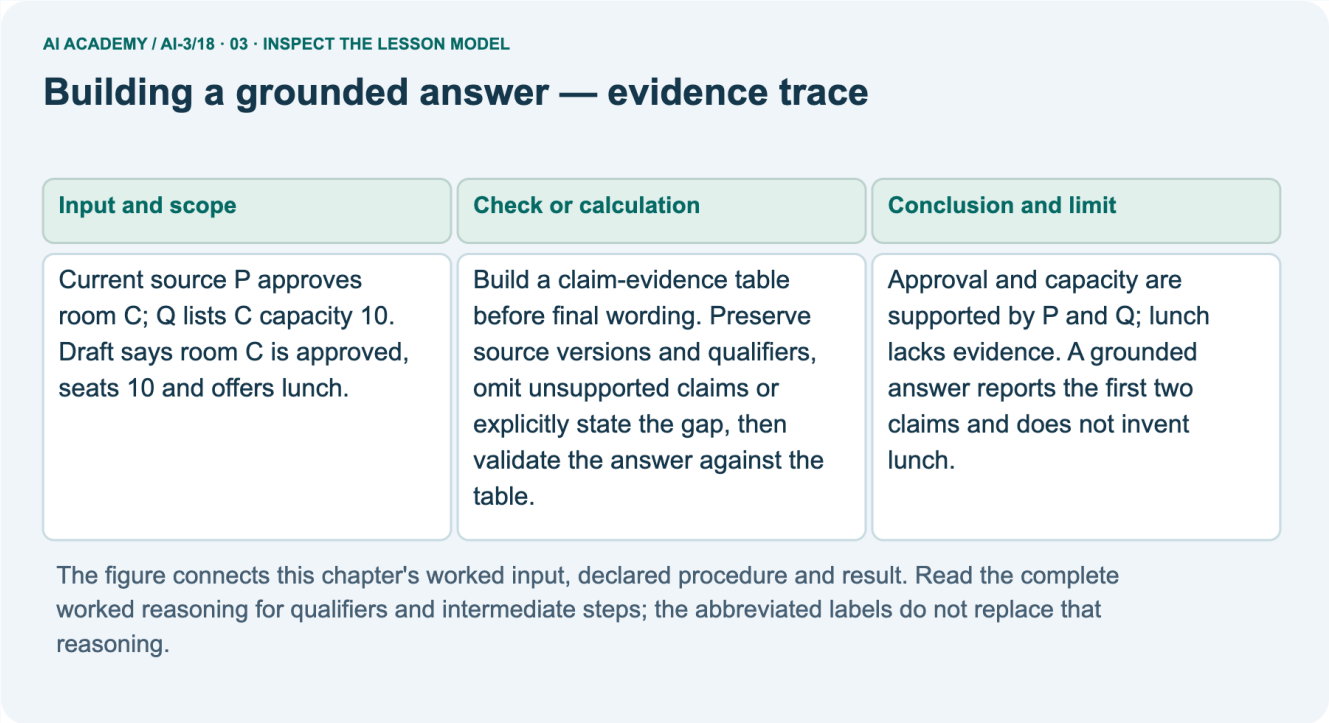
Follow the labelled picture with the matching explanation. Open the vector diagram

Read source: S4 supports the material identity and the solid-to-liquid change. → Draft two claims: Draft: ice is solid water; warming can turn it into liquid water. → Check each claim: Both claims match S4.

A claim that melting takes exactly two minutes would not match.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Building a grounded answer — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

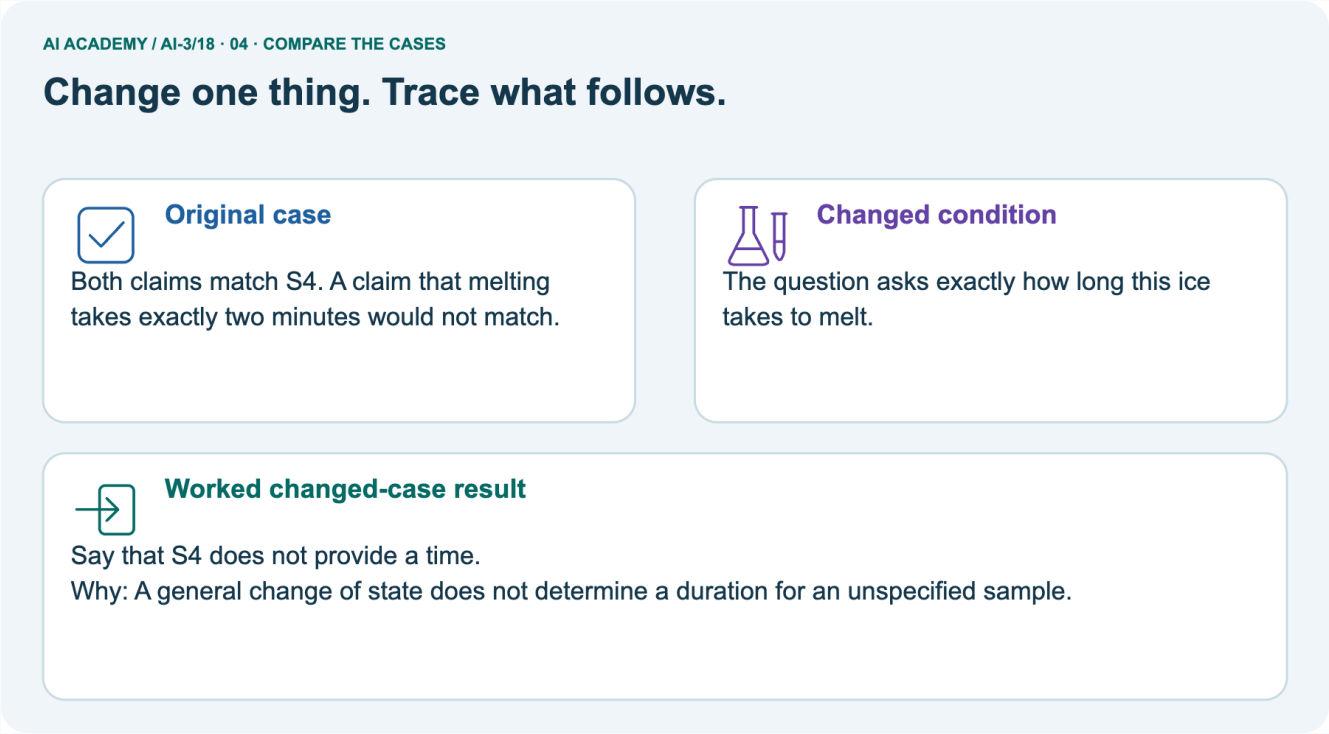
Building a **grounded answer** — **evidence** trace. Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. | Build a claim-**evidence table** before final wording.

Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table. | Approval and capacity are supported by P and Q; lunch lacks **evidence**.

A **grounded answer** reports the first two claims and does not invent lunch.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



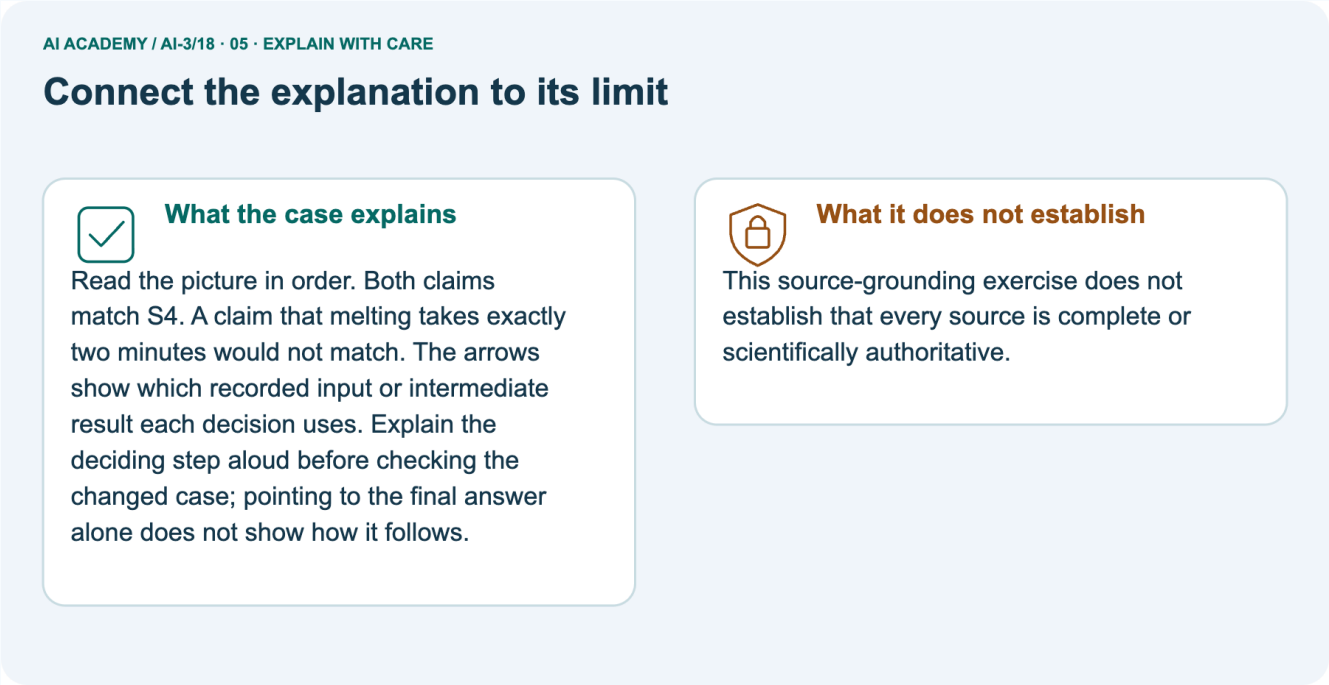
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both claims match S4. A claim that melting takes exactly two minutes would not match. Changed condition: The question asks exactly how long this ice takes to melt.

Expected result: Say that S4 does not provide a time. Reason: A general change of state does not determine a duration for an unspecified sample.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

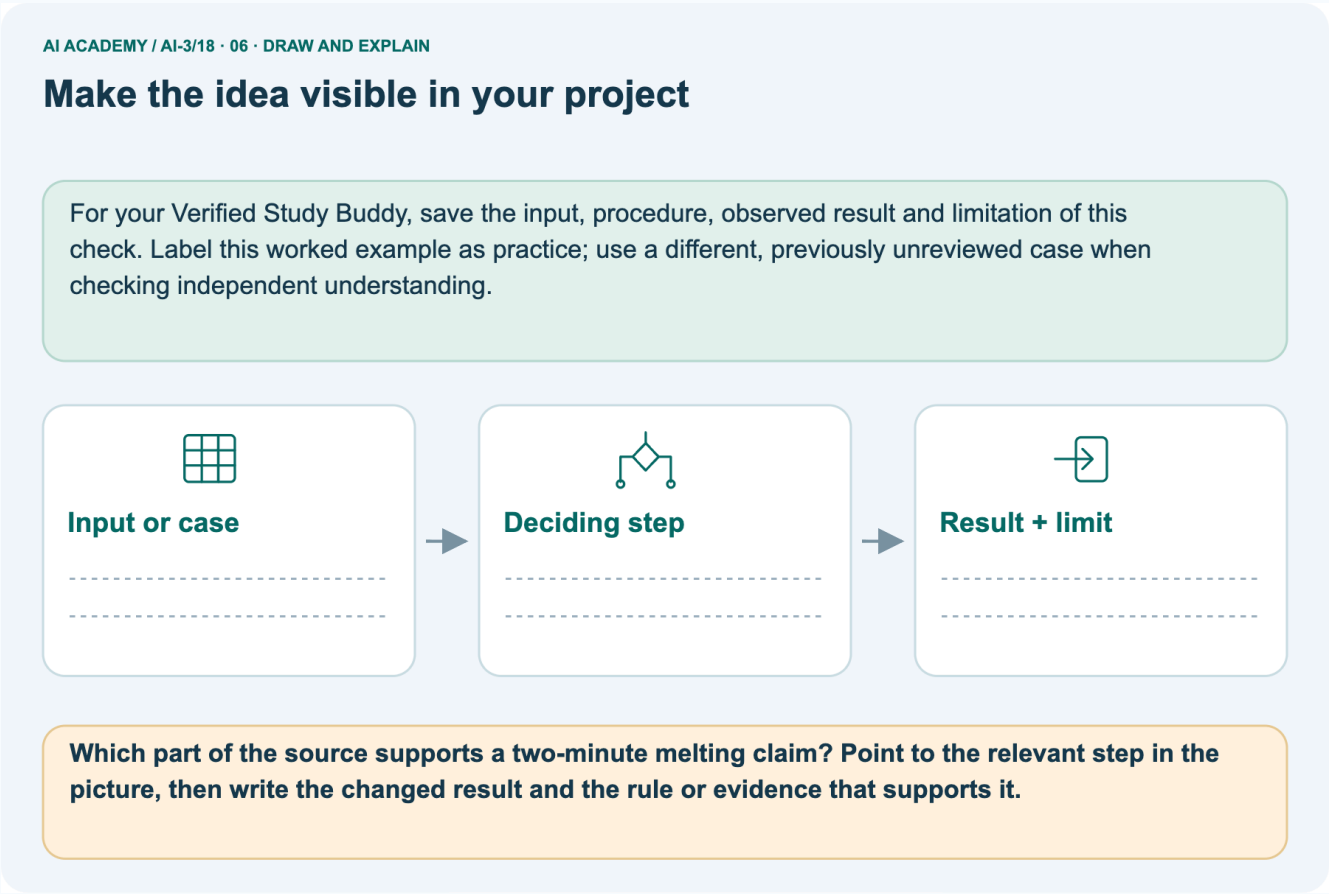
Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.